

Measuring Topical Attention in the Canadian House of Commons, 1920–1949: A Comparison of Induced-Taxonomy LLM Labels and KMeans Clusters

Mathias Bühler

May 2026

Abstract

What does each of two now-standard text-as-data pipelines actually measure when applied to the same parliamentary corpus? I classify 161,633 Canadian House of Commons speeches from 1920 to 1949 with two pipelines. The first asks Google’s Gemini to write a short free-text topic for every speech and then induces a ten-category taxonomy from the 2,000 most frequent topic strings; the second runs MiniBatchKMeans with $k=20$ on a TF-IDF representation of the same speeches. The two pipelines recover the wartime spike with very different sharpness: the LLM-induced *Veterans’ and Military Affairs* category rises from 2.8% in 1935 to 34% in 1942; the closest KMeans clusters (wartime international, defence, wartime labour) sum to roughly one-third that magnitude at peak. The KMeans residual absorbs 37% of mass; the LLM *Government and Parliamentary Affairs* residual absorbs 17% but mixes substantive procedural content with topics the induced taxonomy did not place elsewhere. The two views agree on the timing and direction of every legible historical episode in the period (Bank of Canada Act, Wheat Board, Statute of Westminster, WWII) and disagree on magnitudes in ways that follow mechanically from how each pipeline represents meaning. The methodological note is narrow: KMeans labels by vocabulary co-occurrence, the LLM labels by semantic content, and the gap between the two is the part of topical attention that ordinary words fail to encode.

1 Introduction

Between 1920 and 1949 the Canadian House of Commons debated two wars, a depression, the founding of the Bank of Canada, the Statute of Westminster, the Canadian Wheat Board, and the first national old-age pension. The Hansard record of those debates is dense, long, and unevenly procedural. A researcher who wants to know what Parliament spent its time on, year by year, faces a measurement problem with two now-standard solutions: hand a large language model a category list and let it label every speech, or vectorize the corpus and cluster it without supervision. Both are cheap to run; neither is unambiguous about what it measures.

This paper asks the narrow question: when both pipelines are applied to the same 161,633 speeches with the same year range, what does each one measure, and where do the two answers diverge? The answer is that the two pipelines agree on the timing and direction of every legible historical episode in the period but disagree systematically on the magnitudes. The LLM-induced category for Veterans’ and Military Affairs jumps from 2.8% of speeches in 1935 to 34% in 1942; the three KMeans clusters whose top terms read as wartime (international, defence, wartime labour) combined reach roughly one-third of that share at peak. The Bank of Canada Act window of 1933–1935 shows up as a clean spike in a dedicated KMeans banking cluster (2.3% of speeches at peak); it shows up in the LLM view as a quieter rise inside the Banking and Monetary Policy

category (3.1% of total speech mass across the whole period). The same gap appears in reverse for every wartime episode: the LLM, with access to semantic labels, sees a speech that opens by accepting the budget and proceeds to talk about munitions as a wartime speech; KMeans, with access only to vocabulary, splits that speech across the budget and the wartime cluster according to which words dominate.

The two-pass induced-taxonomy design in the LLM pipeline is the one substantive method choice that needs flagging upfront. Rather than fixing a category schema before classification, the pipeline first asks the model to write a free-text noun phrase describing each speech, producing 129,899 distinct phrases across the corpus. It then derives ten mutually exclusive categories from the 2,000 most frequent phrases and assigns each unique phrase to one category. Measurement and taxonomy are separated: the topic phrase survives any later taxonomic revision, and changing the taxonomy is a two-minute re-run rather than a re-classification of the full corpus. The fixed-taxonomy version of the same pipeline (the baseline I retain for comparison) puts 30–40% of speeches into an *Other* bucket; the induced version pushes that residual down to a *Government and Parliamentary Affairs* category that absorbs 17%. The headline contrast in this paper is therefore not LLM versus KMeans in general, but: an induced-taxonomy LLM versus a TF-IDF KMeans, both run on the same speeches with the same year coverage.

The descriptive payoff is a paired time series of topical attention in the Commons over the inter-war and wartime period. The substantive findings are familiar to anyone who knows Canadian political history: a wartime spike from 1939 to 1945, a Depression-era turn to federal-provincial relations and relief, a sharp banking debate in 1933–1935, agricultural and Wheat-Board volume mid-decade. What is new is that they can be read off two different measurement instruments, and that the gap between the two is informative about what each instrument captures.

The methodological note is narrow. KMeans on bag-of-ngrams clusters by vocabulary co-occurrence, full stop; an LLM with access to a semantic representation of the text clusters by meaning. The wartime spike is the cleanest demonstration: a speech about war that uses budget vocabulary is read as wartime by the LLM and as budget by KMeans. Neither answer is wrong. They are measurements of different things. The implication for downstream work is that unsupervised clustering is appropriate for discovery (what themes are in this corpus at all?), and an LLM classification — preferably with a data-induced rather than researcher-imposed taxonomy — is appropriate for measurement (what share of speeches in 1942 are about the war?). Treating either as a substitute for the other is the failure mode worth naming.

2 Data

The corpus is built from twelve Excel files of speech-level Hansard records covering 1920–1949 (`input/Speeches{1920|1930|1940}_{0..3}.xlsx`). The raw files contain 1.03 million speech rows from the ParlInfo / lipad debates record. The two pipelines apply different filters because they answer different questions.

The LLM pipeline operates on the MP-joined panel produced by an earlier step of the project, `output/speeches_all_mps.csv`, which restricts the corpus to 161,633 MP-attributed speeches. Each row carries speaker name, party, riding, province, year, and the cleaned speech text. Speeches longer than 6,000 characters (well above the 99th percentile of length, which sits near 2,700 characters) are truncated for the model.

The KMeans pipeline operates on a different subset of the same raw files. After dropping duplicate speech text and rows with fewer than 100 whitespace-separated words, the analyzable corpus is 188,583 speeches. The 100-word floor removes 77% of the de-duplicated corpus and is by

far the most consequential filter: without it, the corpus is dominated by procedural interjections with no distinguishing vocabulary, and KMeans collapses 57% of the sample into one undifferentiated residual. The floor is the smallest single change that breaks the residual cluster’s dominance. The KMeans corpus is therefore larger than the LLM corpus (it does not require an MP-attribution join) but more aggressively cleaned for content.

Both pipelines preserve the year as the primary temporal index. Neither uses the speaker’s party as a primary signal: party identification enters the LLM prompt as metadata but not as a category determinant, and it does not enter the KMeans vectorizer at all.

3 Pipeline 1: free-text topics with an induced taxonomy

The LLM pipeline runs in two passes. The first produces a free-text topic per speech; the second induces a ten-category taxonomy from the data and assigns each unique topic to a category. The design rationale is to separate measurement from taxonomy: pass-1 output survives any later change in the category schema, and revising the schema is a two-minute re-run of pass 2 rather than a full re-classification.

Pass 1: free-text topic extraction. Each speech is sent to `gemini-flash-lite-latest` with a prompt containing the year, speaker metadata, the ParlInfo topic hints (`maintopic > subtopic > subsubtopic`), and the truncated raw text. The system instruction asks for a short noun phrase (≤ 8 words, lowercase, no period) describing the substantive topic, with a directive to be specific (“old-age pensions” not “social policy”; “St. Lawrence Seaway construction” not “infrastructure”). No structured-output constraint is imposed; the response is post-processed by stripping quotes and trailing periods. Temperature is 1.0 to encourage specificity, with the consequence that the same speech can in principle receive different phrasings across runs. Async concurrency 80, throughput around 135 requests per second, full panel in roughly 20 minutes. Of the 161,633 speeches, zero returned errors and zero returned empty topics in the production run. The output is 129,899 distinct phrases. The distribution is long-tailed: 80% of phrases appear exactly once, and the top 2,000 phrases cover 12.4% of speeches. This is the central trade-off of free-text measurement — granular but not directly aggregable.

Pass 2, stage 1: induce the taxonomy. The top 2,000 most frequent topic strings, with their frequency counts, are passed in a single call to `gemini-2.5-pro` with a structured-output schema enforcing exactly ten items each containing a category name and a one-sentence definition. The prompt asks for mutually exclusive, collectively exhaustive, empirically grounded categories, stable across the 1920–1949 period, and explicitly instructs the model to derive the categories from what it sees in the data rather than from a generic political-science taxonomy. Temperature is 0 for determinism. The stronger model is justified here because the call is one-shot and the output drives every downstream label. The induced categories and their shares of total speeches are listed in Table 1.

Table 1: Ten induced categories and their share of all 161,633 speeches, 1920–1949. Categories produced in a single call to `gemin-2.5-pro` from the 2,000 most frequent free-text topic strings.

#	Category	Share
1	Government and Parliamentary Affairs	16.7%
2	Transportation and Communications	13.4%
3	Agriculture and Food Policy	11.9%
4	Veterans’ and Military Affairs	11.5%
5	Public Finance and Taxation	10.6%
6	Social Welfare and Labour	9.2%
7	Justice, Constitution, and Indigenous Affairs	9.1%
8	International Trade and Tariffs	9.0%
9	Natural Resources and Intergovernmental Relations	5.7%
10	Banking and Monetary Policy	3.1%

Pass 2, stage 2: assign each topic to a category. Each of the 129,899 unique topic strings is assigned one category. Two efficiency moves keep the call count down. First, the assignment operates on unique topic strings, not on speeches, so the call count is 129k rather than 161k. Second, topics are batched 50 per call into a JSON array with a response schema enforcing a same-length array of category-name enums. About 2,600 API calls total, throughput around 1,360 topics per second, total wall time roughly 1.6 minutes. Temperature is 0; assignments are deterministic conditional on the topic. The per-speech label is then a left-join on the topic field. Two speeches with the same topic string always receive the same category by construction.

Three caveats on the induced taxonomy. First, the *Government and Parliamentary Affairs* category is partly playing the role that an *Other* bucket plays in a fixed-taxonomy run. Its top topics mix substantive procedural matters (electoral redistribution, civil-service salary deductions, House standing orders) with topics that arguably belong elsewhere (continuation of wartime economic controls, National Emergency Transitional Powers Act, establishment of the Department of Reconstruction). At 16.7% of speech mass, it is the largest single category, and any analysis that uses it should expect mixed content.

Second, three themes that a researcher in this period might want to track separately are absorbed into broader categories: *Education* scatters across Social Welfare and Labour (training, vocational programs) and Justice, Constitution, and Indigenous Affairs (residential schools); *Health care* is folded into Social Welfare and Labour; *Immigration* is folded into Justice, Constitution, and Indigenous Affairs alongside citizenship law. A user who needs these as separate dimensions can either re-run induction with a larger sample, hand-edit the taxonomy JSON and re-run the 1.6-minute assignment pass, or post-filter the topic column on keyword matches.

Third, pass 1 uses temperature 1.0 and is therefore non-deterministic at the topic-string level. Pass 2 is deterministic at the category level, conditional on the topic. A re-run of pass 1 would produce slightly different phrasings; the category assignments would shift only at the boundary where the rephrasing changes the dominant noun.

4 Pipeline 2: KMeans on TF–IDF bag-of-ngrams

The KMeans pipeline applies a three-stage clean to the speech text, vectorizes with TF–IDF over unigrams plus bigrams plus trigrams, and fits MiniBatchKMeans with $k=20$. The full pipeline runs in 2 minutes 54 seconds.

Preprocessing. Stage 1 is a cheap per-token pass: lowercase, regex-tokenize on `[a-zA-Z]+`, drop tokens shorter than three characters, drop NLTK English stopwords plus a custom banlist (described below), drop tokens with no WordNet synset whose POS is noun or adjective. Stage 2 collects every surviving un-stemmed token into a vocabulary of about 41,000 unique terms and runs the NLTK Perceptron POS tagger once on the vocabulary, not per speech, keeping terms tagged `NN*` or `JJ*`. Stage 3 drops tokens not in the POS-tagged allowlist and Porter-stems the rest. The two-stage POS design — WordNet pre-filter, then context-free Perceptron tag on the vocabulary — keeps the per-speech work cheap while still using real POS information. Verbs like *inform*, *answer*, *read*, *done*, *agree* survive the WordNet check (they have noun senses) but are correctly dropped by the Perceptron tag.

The banlist is extended in two waves beyond the initial user-supplied list. Procedural noise (*hon*, *sir*, *madam*, *speaker*, *house*, *committee*, *order*, *say*, *said*, *know*, *think*, *shall*, *would*) survives POS filtering as nouns but carries no thematic content in a parliamentary corpus. Aggressive residual-killer additions (*time*, *year*, *day*, *matter*, *case*, *fact*, *point*, *sort*, *kind*, *thing*, *way*, *view*, *opinion*, *regard*, *respect*, *reason*, *occasion*, *instance*, plus *minister(s)*, *people(s)*, *work(s)*, *govern*, *government(s)*, *member(s)*) were the top terms of an early residual cluster that absorbed 45% of speech mass. Banning them forces KMeans to find lexical structure elsewhere. This is a stylistic intervention, not a neutral one: *minister*, *people*, and *government* are content words in a substantive sense, and banning them flattens part of the signal. The justification is that they had become so high-frequency they were drowning out everything else; the trade-off is that the resulting prime-minister cluster and the federal-provincial cluster now identify themselves through other distinctive terms.

Vectorization and clustering. TF-IDF with `ngram_range = (1, 3)`, `min_df = 10`, `max_df = 0.5`, custom token pattern `\S+` (the tokens are already cleaned and space-separated). Output matrix is 188,583 documents by 379,233 features, sparse. MiniBatchKMeans with `k=20`, batch size 4096, `max_iter 200`, `n_init = 5`, `reassignment_ratio = 0.01`, `random_state = 0`. Full KMeans was rejected: at this scale it runs in 30+ minutes per iteration, and the cluster-purity gain at `k=20` is small relative to the iteration cost during banlist tuning.

Tuning sequence. Three runs preceded the final output, and the failures matter as much as the success. Run 1 (`k=10`, unigrams plus bigrams, no length filter) produced a 57% residual dominated by verbs — the WordNet-only POS filter was too permissive. Run 2 (same plus 100-word length filter and Perceptron POS pass) cut the residual to 45%. Run 3 (`k=20`, bigrams plus trigrams only, aggressive banlist) collapsed: 76% of mass landed in one cluster and eight others became singletons. Removing unigrams in favor of bigrams plus trigrams seems intuitive (“we want compound topics like *wheat board*”) but the dense low-rank signal in unigrams is what keeps document vectors from collapsing onto each other in the sparser ngram space. Run 4 (`k=20`, unigrams plus bigrams plus trigrams, `min_df = 10`, aggressive banlist, 100-word floor) is the final output: largest cluster at 37% of mass, all 20 clusters non-degenerate. Top-15 stems per cluster, on which the labels in Table 2 are based, are persisted in `output/cluster_top_terms.2605.csv`.

Table 2: KMeans cluster labels, sizes, and representative top stems (Porter-stemmed). “Residual” marks the one cluster whose top terms have no coherent theme. Two pairs (wheat: 4/8; railways: 6/19) are partial duplicates: KMeans split each theme across two centroids based on subtle vocabulary differences.

#	Label	Share	Selected top stems
0	Elections / electoral machinery	1.1%	chief elector offic, voter, candid
1	Taxation (income, tariff, budget)	4.3%	tax, incom, incom tax, tariff, sale tax
2	Residual	36.7%	canada, depart, act, countri
3	Post office	2.9%	postmast gener, mail, tender, contractor
4	Wheat board	1.2%	canadian wheat board, bushel, grain
5	Wartime labour / manpower	6.9%	men, board, servic, forc, employ
6	Railways (CN / CP)	2.2%	canadian nation, pacif, canadian pacif railway
7	Party politics	4.2%	parti, leader, opposit, conserv, liber
8	Farming / western Canada	2.9%	farmer, wheat, grain, western canada
9	Wartime international	5.1%	war, world, peace, britain, forc
10	Prices / trade / markets	7.9%	price, trade, market, product
11	Maritime provinces	1.8%	new brunswick, princ island, nova
12	Old-age pensions	2.0%	old age pension, pension act, veteran
13	Finance / insurance / companies	3.3%	compani, insur, profit, stock, rate
14	Civil service	1.5%	civil servic commiss, salari, employe
15	Legislation procedure	5.3%	bill, amend, read, second read, motion
16	Banking	1.2%	bank canada, credit, gold, central bank
17	Defence	1.6%	defenc, nation defenc, militari, naval
18	Federal-provincial relations	4.6%	provinc, dominion, feder, municip, relief
19	Railways (BC / Pacific)	3.4%	british columbia, pacif, canadian, line

Nineteen of the twenty clusters are thematically coherent. Cluster 2 (the residual, top terms *canada*, *depart*, *act*, *countri*) holds 37% of the corpus. Two pairs are partial duplicates: wheat appears in cluster 4 (with the *wheat board* ngram dominant) and cluster 8 (with *western canada*, *farmer*, *crop*); railways appears in cluster 6 (Canadian National / Pacific) and cluster 19 (British Columbia / Pacific lines). This is a known KMeans pathology — the algorithm has no incentive to merge centroids that occupy distinct regions of the vocabulary space even when they cover the same theme — and it survives in the final output.

5 Comparison

I now place the two pipelines side by side. The two output series are not directly comparable in level: the LLM ten-category schema necessarily allocates more mass per category than the KMeans twenty-cluster schema. What is comparable is the timing and direction of any year-over-year shift and the share of mass that ends up in an unlabeled residual.

5.1 The wartime spike

The cleanest test of the gap between the two pipelines is the WWII window. The LLM *Veterans’ and Military Affairs* share by year is reproduced in Table 3.

Table 3: Share of speeches assigned to *Veterans’ and Military Affairs* by year, induced-taxonomy LLM pipeline. The model never sees the year as a primary signal for category assignment: year enters the pass-1 prompt as metadata, and pass-2 sees only the topic string.

1925	1930	1935	1939	1940	1942	1944	1945	1947	1949
7.3%	6.6%	2.8%	14.7%	18.1%	34.0%	23.6%	29.1%	15.2%	7.7%

The share rises from 2.8% in 1935 to 34% in 1942 and falls to 7.7% by 1949. WWI demobilization and pension administration carry the pre-1935 baseline; the 1939–45 spike captures conscription, the Bren machine-gun contract investigation, war powers continuation, and the post-war Soldier Settlement Act revaluations. The taxonomy recovers a well-known historical pattern from text and metadata alone, with no temporal training signal in the assignment step.

The closest KMeans analog is the sum of three clusters whose top terms read as wartime: wartime international (cluster 9, 5.1% of corpus), defence (cluster 17, 1.6%), and wartime labour (cluster 5, 6.9%). Summed across the corpus these three clusters carry 13.6% of mass; at the 1942 peak they reach roughly 12% of yearly speeches in the stacked-area plot (Figure 2). That is roughly one-third the LLM magnitude.

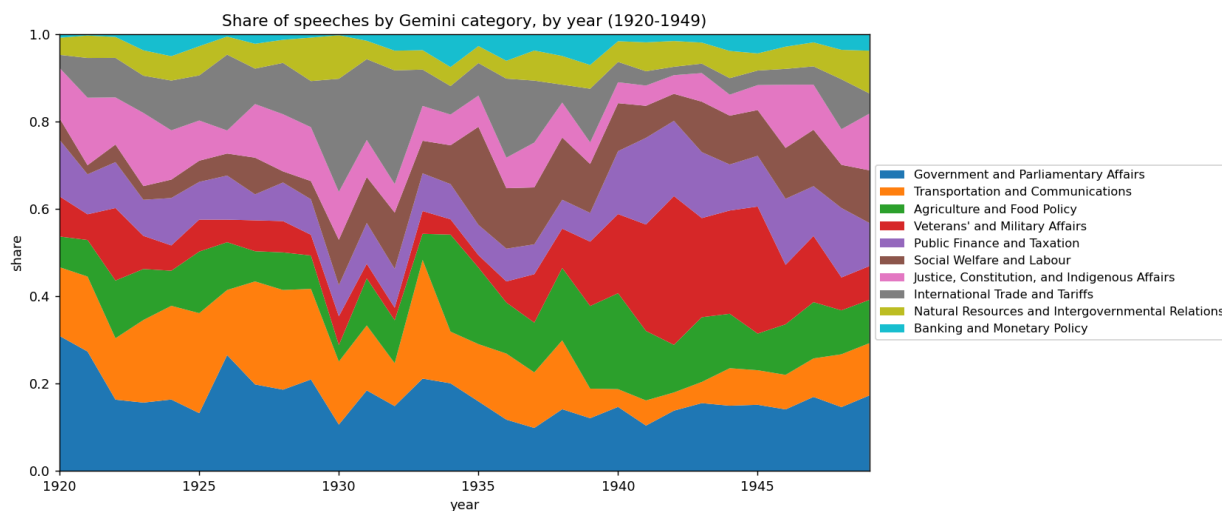


Figure 1: LLM induced-taxonomy category shares by year, 1920–1949 (Pipeline 1). Each year sums to 1 across the ten categories. The *Veterans' and Military Affairs* band rises sharply 1939–1945.

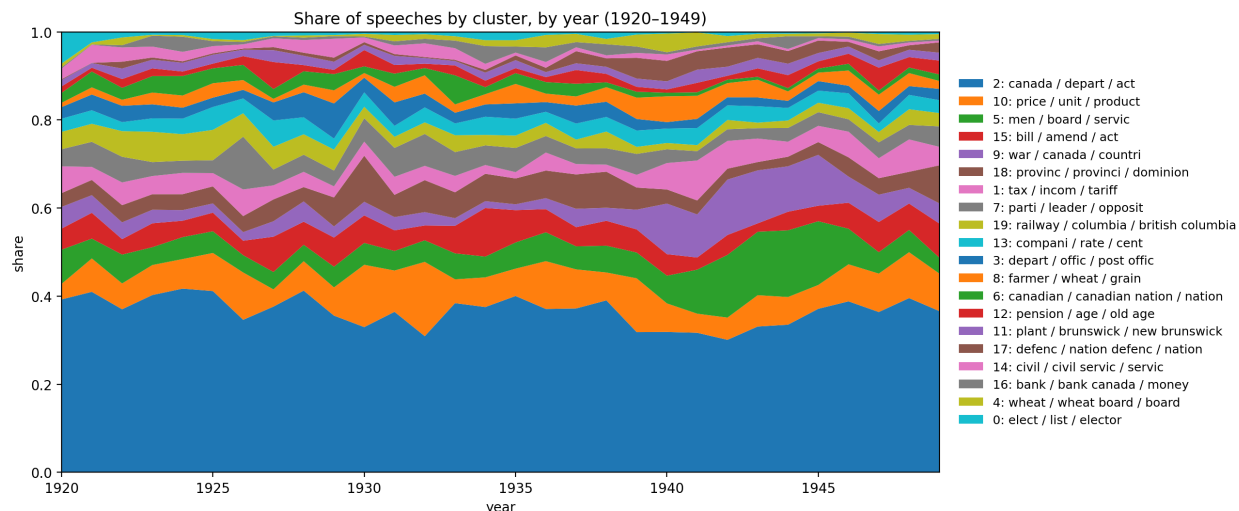


Figure 2: KMeans cluster shares by year, stacked (Pipeline 2). Each year sums to 1 across the twenty clusters. The wartime bands (defence, wartime international, wartime labour) rise 1939–45 but with much smaller magnitude than the LLM equivalent.

The gap is not a tuning failure. It is built into what the two pipelines measure. The LLM reads a speech that says “we must defend Canada,” identifies the topic as the war effort, and assigns the speech to *Veterans’ and Military Affairs* regardless of whether the speech also contains tax vocabulary, budget vocabulary, or trade vocabulary. KMeans reads the same speech, projects it into a 379k-dimensional sparse TF-IDF space, and assigns it to whatever cluster centroid is nearest in that space — which will usually be the budget or trade cluster, because budget and trade speeches dominate the corpus by volume and many wartime speeches happen to use a lot of budget and trade words. The LLM has access to a semantic representation; KMeans has access only to lexical co-occurrence.

This is the methodological note. It generalizes: any episode in which the substantive topic of speeches differs from the vocabulary that those speeches share with the rest of the corpus will show up sharply in the LLM pipeline and weakly in the KMeans pipeline. The wartime spike is the largest such episode in this period, which is why it makes the cleanest test.

5.2 Other temporal episodes

Several smaller episodes line up between the two pipelines and provide a check on whether the LLM is finding patterns that any reasonable measurement instrument would find, or whether it is hallucinating structure that vocabulary cannot see.

Bank of Canada Act, 1933–1935. The KMeans banking cluster (cluster 16, 1.2% of corpus) peaks sharply in 1933–1935 in the facet plot (Figure 3). The LLM *Banking and Monetary Policy* category (3.1% of corpus) shows the same window with the same shape. Both pipelines pick up the legislative event with no temporal supervision.

Federal-provincial relations, 1930–1935. The KMeans federal-provincial cluster (cluster 18, 4.6%) peaks 1930–35, picking up the Statute of Westminster (1931), the Bennett government’s relief debates, and early federal-provincial transfers under the Depression. The LLM *Natural Resources*

and *Intergovernmental Relations* category (5.7%) shows a compatible peak in the same window. Two different operational definitions of the same theme produce the same temporal signature.

Wheat Board, 1935. The KMeans wheat-board cluster (cluster 4) and farming/western-Canada cluster (cluster 8) both spike mid-1930s, consistent with the Canadian Wheat Board Act (1935) and the Prairie Farm Rehabilitation Act (1935). The LLM *Agriculture and Food Policy* category (11.9%) shows the corresponding peak. The duplication of the theme across two KMeans clusters — one keyed on the wheat-board ngrams, one on western-Canada vocabulary — is the KMeans pathology again and is hidden in the LLM aggregate.

Elections, 1921 and 1925. The KMeans elections cluster (cluster 0, 1.1%) peaks 1921–1925, matching the 1921 federal election and the 1925 minority parliament. The LLM *Government and Parliamentary Affairs* category, which absorbs election-machinery topics among other procedural content, peaks in the same window but with a much higher baseline because of its residual character.



Figure 3: KMeans cluster share of yearly speeches, faceted. Each panel is one cluster on its own vertical scale. The legible temporal features (banking 1933–35, federal-provincial 1930–35, wheat mid-1930s, wartime clusters 1939–45) appear as sharp spikes; the residual cluster (panel 2) is large and flat.

5.3 The residual problem in two forms

Both pipelines have a residual category. The KMeans residual (cluster 2, top terms *canada*, *depart*, *act*, *countri*) absorbs 37% of speech mass; the LLM *Government and Parliamentary Affairs* category absorbs 17%. These are not the same kind of object.

The KMeans residual is what the algorithm produces when document vectors are dense but undifferentiated. Its top terms are the most generic content words that survive the aggressive banlist — words that appear in many speeches without distinguishing any thematic substructure. It is the cluster that absorbs speeches whose vocabulary is mostly the corpus average. There is no way to interrogate its contents for substantive topics short of running a second clustering pass within the residual.

The LLM *Government and Parliamentary Affairs* residual is different. Its top topic strings are interpretable individually: electoral redistribution, civil-service salary deductions, House standing orders, continuation of wartime economic controls, National Emergency Transitional Powers Act, establishment of the Department of Reconstruction. Some of these are substantive procedural topics; some are wartime topics that the induced taxonomy could not place into *Veterans' and Military Affairs* because the topic string did not mention the war directly. The 17% residual is therefore a mixed object: roughly half genuine procedural content, roughly half topics that a slightly better taxonomy would route elsewhere.

The implication for downstream use is concrete. A researcher building a yearly time series of, say, social-policy attention can use the LLM *Social Welfare and Labour* category as a measurement instrument and expect modest measurement error from the residual contamination. A researcher building the same series from KMeans clusters has to either accept that 37% of the corpus is uncategorized, or do extra work to recover the residual.

6 Discussion

What each pipeline measures, in one sentence each. KMeans on TF-IDF measures vocabulary co-occurrence at the document level: two speeches are close if they use similar words. The induced-taxonomy LLM measures topical substance: two speeches are close if a reader would describe them with the same noun phrase. The two definitions agree when topical substance is well-encoded by ordinary vocabulary (banking, the Wheat Board, federal-provincial relations) and disagree when topical substance crosses vocabulary boundaries (the war effort, which appears in budget speeches, trade speeches, and procedural speeches simultaneously).

Reproducibility. The LLM pipeline depends on a deterministic config for pass 2 (temperature 0) and a non-deterministic config for pass 1 (temperature 1.0). Pass-1 re-runs will produce different topic strings; pass-2 assignments are deterministic conditional on the topic. A re-run that wanted full determinism could drop pass-1 temperature to 0 at the cost of less varied topic phrasing. The KMeans pipeline is fully deterministic given the random seed: identical inputs and parameters reproduce identical assignments. Determinism cuts both ways. The KMeans output is reproducible bit-for-bit by anyone with the data; the LLM output requires a working API key, a small but non-zero dollar cost (the full panel costs roughly five dollars in Gemini Flash credits at current pricing), and a stable model version.

Cost. The KMeans pipeline runs in 2 minutes 54 seconds on a laptop. The LLM pipeline runs in roughly 22 minutes end-to-end: 20 minutes for pass 1 (async concurrency 80, throughput around 135 requests per second), 30 seconds for the one pass-2 induction call, 1.6 minutes for pass-2 assignment. The cost differential is therefore an order of magnitude in wall time and several dollars in API spend. For exploratory work on a single corpus, the KMeans budget is easier to justify; for repeated measurement on a moving target (new speeches every year, new questions every quarter), the LLM cost amortizes well.

Recommendation. The two pipelines are complements, not substitutes. KMeans is appropriate for first-pass discovery: what themes are in this corpus at all, and which ones have enough distinctive vocabulary to surface as clusters? The induced-taxonomy LLM is appropriate for measurement: what share of speeches in 1942 are about the war? Treating either as a substitute for the other — using KMeans cluster sizes as a measurement of topical attention, or skipping the discovery step and trusting an LLM to know what categories the corpus contains — is the failure mode I would warn a downstream user against.

Limitations. The KMeans pipeline operates on 188,583 speeches after the 100-word filter; the LLM pipeline operates on 161,633 MP-attributed speeches without a length filter. The two corpora overlap heavily but are not identical, which prevents a speech-by-speech alignment table. The KMeans residual at 37% and the LLM residual at 17% are not strictly comparable in level because the underlying samples differ. The paper does not validate either pipeline against a hand-coded gold standard, benchmark against alternative LLMs or unsupervised methods, or test whether LLM topic assignments are stable across re-runs at temperature 1.0. None of these changes the headline comparison, but each would need to be addressed before either output could serve as the measurement instrument in a downstream causal design.